2.	Motor is not running at rated speed.	1. Low voltage	Check the wiring
3.	When motor is started, the fuse gets melted.	1. Over load.	Reduce the load.
		2. Short-circuit in motor circuit.	Check the short circuit and rectify it.
4.	Motor stops when started.	1. Major fault in motor	Replace the motor.
		2. Over load	Reduce the load.
5.	Motor is getting heat when runs with load.	1. Dust in air passage	Clean the air passage and check the outgoing hot air.
		2. Grounded wiring	Check and rectify the fault.
		3. Over voltage	Motor should be operated only with rated voltage.
		4. Rotor touches with the stator	Check alignment and correct it.
6.	Vibration of electrical motors	1. Motor is not fixed properly	Fix it properly.
		2. Fault in bearings	Replace the bearings.
		3. No proper alignment of bearings	Adjust the alignment of bearings.
		4. No proper end play	Replace the washers.
7.	Bearings gets heat when motor rotates.	1. Dust in oil ways of bearings	Clean the oil ways and replace it with new oil.
		2. Damaged bearings	Replace the bearings
		3. Low (or) high viscosity of lubricant.	Apply new lubricant.
		4. Lubricantion is not enough.	Enough lubricantion should be applied.
		5. Higher end thrust.	Rectify the fault.
8.	Rotor rotates in opposite directions.	1. Wrong winding connection	Make the proper connection.

10.5 Testing of new machines

It is important to note the following points.

a. In a motor, check whether the terminal connections and wiring are correct or not as in the wiring diagram.
b. Open the terminal box once in six months to test the terminals for tightening
c. Measure the insulation resistance value.
d. Do the continuity test.
e. The protective device which is used to control the motor is fixed at the 'minimum current value' and 'minimum time setting' before the motor is used.
f. If the rotor rotates in the opposite direction, switch off the supply and change any two connections of the wire. Check whether the rotor is rotating in same direction.

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